

# CS215 Assignment-1

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## 1 Solution Q5

Assume that the IDs of the traders are randomly distributed.

Let the probability of me winning if I choose my position to be  $r$  be  $F(r)$ .

$$F(r) = P(\text{I win} \mid \text{I am at position } r) = P(\text{Traders at } 1, 2, \dots, r-1 \text{ lose}) \times P(\text{I win at position } r)$$

$$F(r) = \begin{cases} 0 & \text{if } r = 1, \\ \frac{200}{200} \cdot \frac{199}{200} \cdot \dots \cdot \frac{200-(r-2)}{200} \cdot \frac{r-1}{200} & \text{if } 2 \leq r \leq 200, \\ 0 & \text{if } r > 200. \end{cases}$$

Now, suppose my winning probability is maximum at  $r = r_0$ .

Then,  $F(r) > F(r-1)$  and  $F(r) > F(r+1)$ .

$$\begin{aligned} \frac{200}{200} \cdot \dots \cdot \frac{203-r}{200} \cdot \frac{202-r}{200} \cdot \frac{r-1}{200} &> \frac{200}{200} \cdot \dots \cdot \frac{203-r}{200} \cdot \frac{r-2}{200} \\ \frac{200}{200} \cdot \dots \cdot \frac{202-r}{200} \cdot \frac{r-1}{200} &> \frac{200}{200} \cdot \dots \cdot \frac{202-r}{200} \cdot \frac{201-r}{200} \cdot \frac{r}{200} \end{aligned}$$

$$\begin{aligned} \frac{202-r}{200} \cdot (r-1) &> (r-2) & \frac{201-r}{200} \cdot r &< r-1 \\ \Rightarrow r^2 - 203r + 202 + 200r - 400 &< 0 & \Rightarrow r^2 - 201r + 200r - 200 &> 0 \\ \Rightarrow r^2 - 3r - 198 &< 0 & \Rightarrow r^2 - r - 200 &> 0 \\ \Rightarrow r \in (-12.65, 15.65) & & \Rightarrow r \in (-\infty, -13.65) \cup (14.65, \infty) \end{aligned}$$

Hence, by taking the intersection of the above intervals, I conclude that the probability of me winning is maximum at  $r = 15$ .